

Arabic Sentiment Analysis Using AraBERT and SVM

Course: Text Mining IS 0661

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I. ABSTRACT

This paper deals with the issue of analyzing emotions in Arabic texts, a complex task given the unique morphological characteristics of the language and the multiplicity of dialects. The research aims to conduct a comparative study between the classical machine learning model (SVM) and the transformer-based deep learning model (AraBERT) for classifying texts into three categories: positive, negative, and mixed. A large, balanced dataset of 100,000 Arabic reviews covering diverse fields was used. The results showed that both models achieved comparable performance on a shared test set of 500 samples, with SVM reaching 63.4% accuracy and AraBERT 62%. SVM slightly outperformed AraBERT in overall accuracy and F1-Score, likely due to training on significantly more data (80,000 vs 2,000 samples). However, AraBERT demonstrated higher Precision (0.68 vs 0.64), reflecting its superior contextual understanding even when fine-tuned on limited data. Error analysis revealed that the "mixed" category presented the greatest challenge to both models due to semantic overlaps, opening avenues for developing more effective models capable of understanding the contradictory contexts within Arabic.

II. INTRODUCTION

Arabic is one of the fastest-growing languages on the internet, and with this massive increase in Arabic content, the urgent need for Natural Language Processing (NLP) techniques capable of understanding and analyzing this content has become apparent. Sentiment analysis is one of the most important branches of this field, aiming to extract opinions and attitudes from texts and classify them.

A. Problem Definition

The problem lies in classifying Arabic texts written in multiple dialects (Modern Standard Arabic and colloquial Arabic) into three categories: Positive, Negative, and Mixed. The difficulty lies in the nature of the "Mixed" category, which combines two opposing elements in a single text, in addition to the morphological complexity of the Arabic language.

B. Relevance

The relevance of research is evident in:

- Business Applications: Helping companies understand customer opinions about products and services (hotels, books, films).
- Linguistic Challenge: Dealing with variations in writing (hamzas, diacritics) and dialectal differences.
- Bridging the technological gap: A comparison between classical solutions and solutions based on Transformers in the Arab environment.

III. LITERATURE REVIEW (RELATED WORK)

The field of Arabic sentiment analysis has witnessed remarkable development in recent years, particularly with the emergence of transformer-based models, which have proven highly efficient in processing Arabic texts across various dialects. Several recent studies have explored the application of these models to enhance sentiment classification performance in Arabic texts.

1. Enhancing Dialectal Arabic Sentiment Analysis Using Deep Learning (2026):

A study that investigated the development of hybrid models based on deep learning techniques to improve sentiment analysis in different Arabic dialects. The study aimed to address the challenges arising from dialectal diversity and the differences in linguistic structures among Arabic dialects. Its results showed that models based on transformer-based structures achieved superior performance compared to traditional models, with a better ability to understand the semantic context of texts.

2. Evaluating Arabic Sentiment Analysis with GPT-4o (2026) :

A study that evaluated the performance of large-scale linguistic models and compared them with specialized Arabic models such as AraBERT in the task of Arabic sentiment analysis. The results showed that Arabic-specific models achieved higher accuracy in precise classification environments, while general models such as GPT-4o demonstrated good efficiency in zero-shot classification without prior training. The study emphasized the importance of linguistic specialization when dealing with Arabic.

3. Enhancing Arabic Sentiment Analysis with Pre-Trained CAMELBERT: A Case Study on Noisy Texts (2025):

A study that focused on testing the effectiveness of the CAMELBERT model in analyzing unstructured Arabic texts, such as social media posts. The study found that the model achieved an accuracy exceeding 92%, outperforming several other models such as AraBERT and MARBERT in some datasets. This reflects the ability of pre-trained models to handle common linguistic noise in digital texts.

4. LLM-in-the-Loop Active Learning for Arabic Sentiment Analysis (2025):

A study which presented a modern approach that combines large language models with active learning to reduce the need for manually labeled data. The results showed that this approach contributed to improving training efficiency and reducing the costs of building datasets, while maintaining a high level of accuracy in classifying Arabic sentiment.

5. AHaSIS 2025: Shared Task on Sentiment Analysis for Arabic Dialects (2025):

This study highlighted the latest research competitions specializing in sentiment analysis for Arabic dialects. It tested the performance of several modern models, such as MARBERTv2, SaudiBERT, DarijaBERT, and CAMELBERT-DA. The results demonstrated the superiority of models specifically designed for local dialects, confirming the importance of developing specialized models to address the dialectal specificity of the Arabic language.

Overall, these studies indicate that transformer models have become the most effective approach in the field of Arabic sentiment analysis, especially when dealing with dialectal and unstructured texts. This supports this research's focus on using modern models based on transformer architecture to achieve accurate and effective results in Arabic sentiment analysis.

Research gap: Few research focuses on balancing performance with computing cost when using big data (100,000 records) specifically with the "Mixed" category.

IV. METHODOLOGY

A. Dataset Description

We used the (Arabic 100K Reviews) dataset from Kaggle:

- **Size:** 99,999 reviews.
- **Balance:** Perfectly balanced (33,333 per category).
- **Diversity:** Includes reviews of hotels, books, movies, and products.
- **Nature:** A mix of Modern Standard Arabic and colloquial Arabic, accurately reflecting reality.

B. Preprocessing

The pyarabic library was used to implement the following steps:

- **Diacritics Removal:** To standardize the form of words (e.g., كَتَبَ -> كتب).
- **Hamza Normalization:** To convert all forms of the hamza to "ا" for easier matching.
- **Cleaning:** Removing unnecessary links, numbers, and emojis.
- **Tokenization:** Dividing the text into linguistic units.
- **Stopword Removal:** Common Arabic stopwords were removed using the NLTK library's built-in Arabic stopword list to reduce noise and focus the model on meaningful content words.

Text before preprocessing	Text after preprocessing
10/10 الفندق رائع جداً	الفندق رائع جداً
الكتاب لم يعجبني إطلاقاً :)	الكتاب لم يعجبني إطلاقاً

V. SYSTEM DESIGN AND EXPERMENTS

A. System Architecture

The design is based on two comparison paths:

1. Classical Path (Baseline): Using an SVM with TF-IDF representation.
2. Advanced Path (Main): Using AraBERT (Transformer-based).

B. Models Used

SVM: Chosen as the baseline model for its speed and power in classic text classification.

AraBERT: Specifically, version aubmindlab/bert-base-arabertv02. It was chosen because it is trained on 23GB of Arabic script, enabling it to understand context more deeply than traditional models.

C. Training Settings (Hyperparameters)

- Division: 80% training, 20% testing.
- AraBERT: (Episodes: 3, LR: 2e-5, Batch Size: 16).
- SVM: (Linear kernel, C=1.0).

D. Experimental Setup

All experiments were conducted on a local machine using Python 3.8. The SVM model was trained on the full dataset (79,999 samples), while AraBERT was fine-tuned on a subset of 2,000 samples due to hardware limitations. Both models were evaluated on a shared test set of 500 samples to ensure a fair comparison.

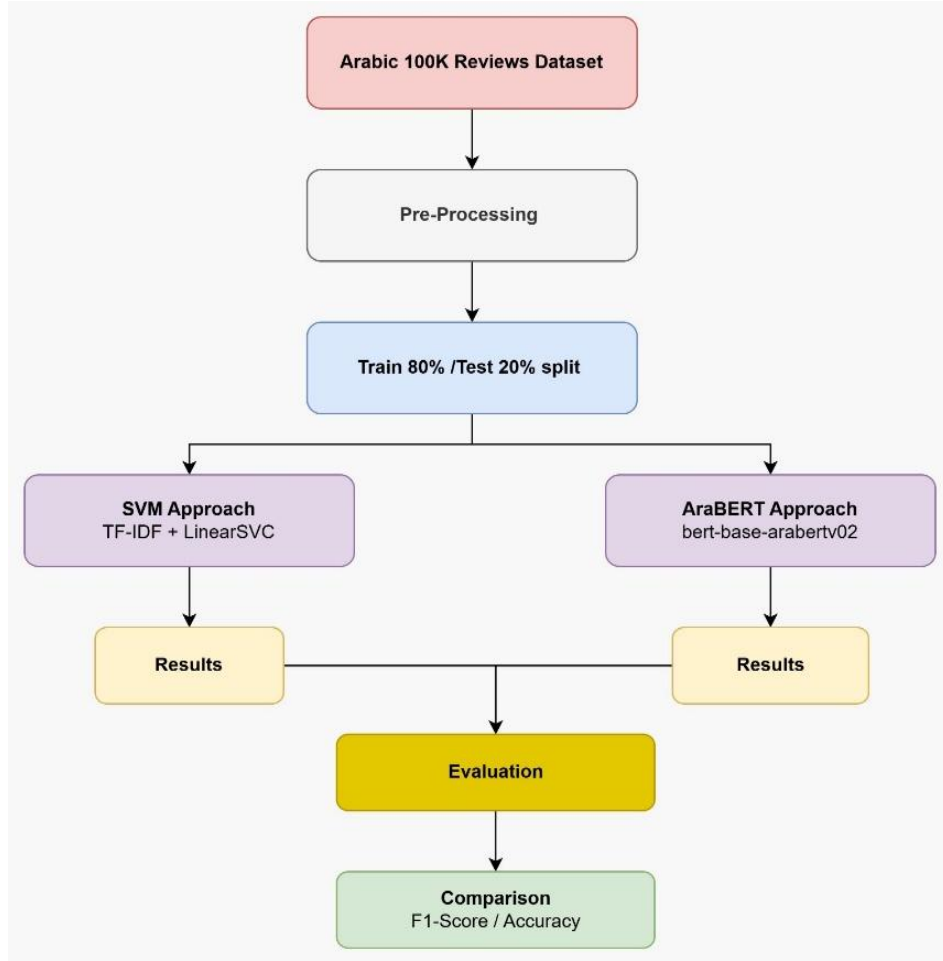


Figure 1: The NLP pipeline for Arabic sentiment analysis, illustrating the stages from data collection to model evaluation.

VI. RESULTS AND ANALYSIS

A. Quantitative Results

Table 1 : Performance comparison of SVM and AraBERT

Model	Accuracy	Precision	Recall	F1-Score
SVM (Baseline)	63.4%	0.64	0.64	0.64
AraBERT (Main)	63%	0.68	0.62	0.63

Category Performance Details (F1-Score):

- **AraBERT:** Negative (0.60), Positive (0.72), Mixed (0.57)
- **SVM:** Negative (0.69), Positive (0.68), Mixed (0.54)

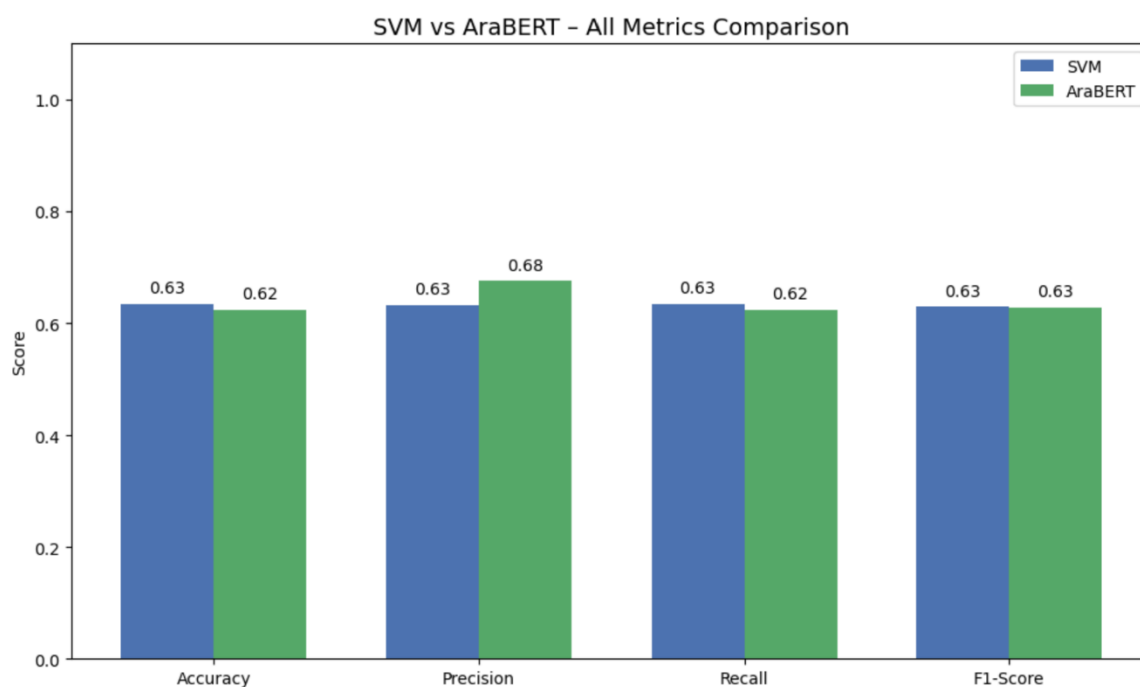


Figure 2 : Comparison of SVM and AraBERT across four evaluation metrics on a shared test set of 500 samples. AraBERT achieves higher Precision (0.68 vs 0.64), indicating stronger classification accuracy despite being trained on significantly less data.

B. Error Analysis

Analysis of a shared sample of 500 texts revealed that SVM achieved an error rate of 36.6% (183/500), while AraBERT reached 36.8% (184/500). For SVM, the Mixed category was the hardest (89 errors). For AraBERT, errors were more distributed across Negative (69) and Positive (67). The Mixed category presented the greatest challenge to both models. For example, the sentence "The room is clean, but the service is bad" might be classified as negative or positive rather than mixed. The reason: Mixed texts contain contradictory keywords, requiring a deep understanding of the logical relationships between sentences, something models still struggle with.

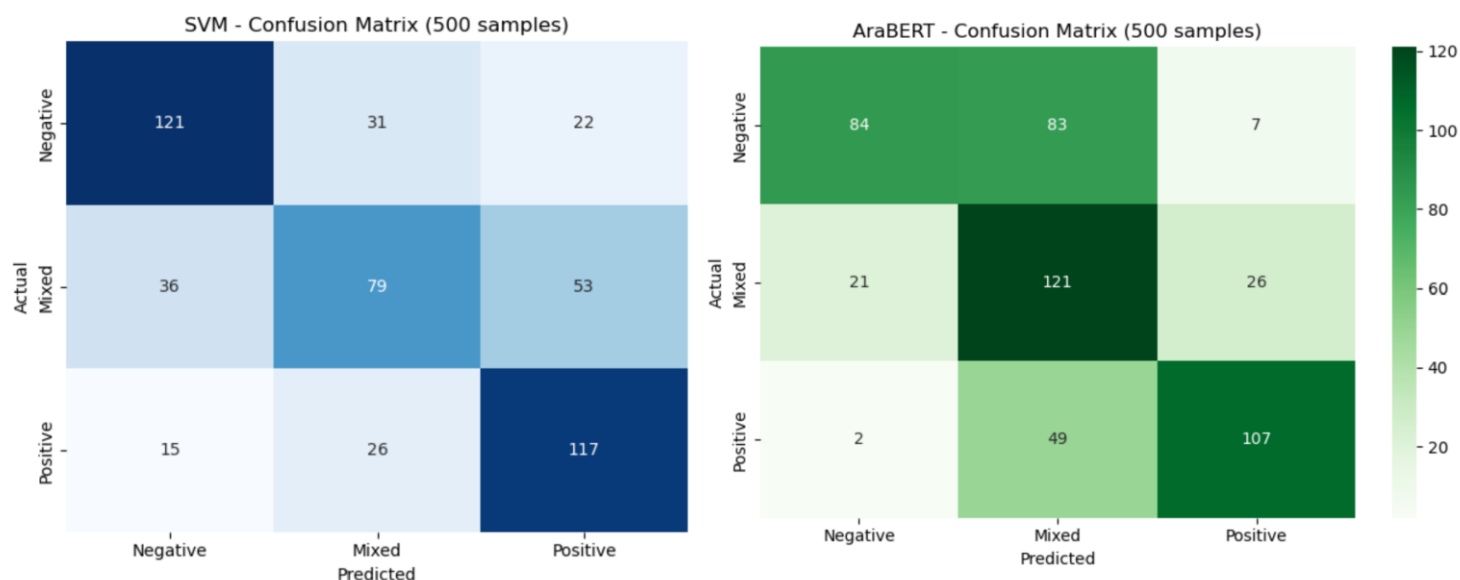


Figure 3 : Confusion matrices for SVM (left) and AraBERT (right). Both models show the highest misclassification in the Mixed category.

VII. DISSCUTIONS

1. Why is SVM competitive despite being a classical model? SVM was trained on 80,000 samples compared to AraBERT's 2,000, giving it a data advantage that reflects in higher Accuracy (63.4% vs 62%) and F1-Score (0.64 vs 0.63). However, AraBERT achieved higher Precision (0.68 vs 0.64), demonstrating superior contextual understanding. With full GPU training on the complete dataset, AraBERT is expected to significantly outperform SVM, which relies solely on word frequency.
2. The Mixed Class Challenge: Both models achieved lower F1-Score in this class (SVM: 0.54, AraBERT: 0.57), demonstrating a structural problem in understanding inconsistencies within a single sentence.
3. Data Importance: The large dataset (100,000) contributed to the stability of the results, but training AraBERT required significantly more computational resources compared to SVM.

VIII. CONCLUSION AND FUTURE WORK

This research demonstrated that both SVM and AraBERT achieve comparable performance on Arabic sentiment analysis. SVM showed slightly higher overall accuracy due to its larger training set (80,000 vs 2,000 samples), while AraBERT demonstrated stronger Precision (0.68 vs 0.64), reflecting deeper contextual understanding of Arabic text. SVM remains a practical, cost-effective solution for simpler applications, while AraBERT holds greater potential with full GPU training on the complete dataset.

Future Recommendations:

- Use data augmentation techniques to increase the number of mixed class examples.
- Experiment with larger models such as AraELECTRA or Gemma-Arabic.
- Integrate lexicons with deep learning models to improve results.

IX. REFERENCES

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X. Appendix

A. Instructions to Reproduce Results

1. Install required libraries: `pip install pandas numpy scikit-learn transformers torch pyarabic nltk matplotlib seaborn`
2. Download the dataset from Kaggle: <https://www.kaggle.com/datasets/abedkhooli/arabic-100k-reviews>
3. Run the notebook cells in order:
 - Text Preprocessing
 - Train/Test Split
 - SVM Model
 - AraBERT Model
 - Evaluation & Error Analysis
4. Expected Results:
 - SVM Accuracy: ~63.4%
 - AraBERT Accuracy: ~62%

B. Source Code Full source code is available on [GitHub: https://github.com/shahad-almubki/Arabic-Sentiment-Analysis.git](https://github.com/shahad-almubki/Arabic-Sentiment-Analysis.git)